Wall time per step

Grey: the step where the run died. Blank: step not available or not reached.

100k (69,412 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	11	1.0		0.4	1.5	4.9	1.7	1.4	2.5	26		8.5
bixverse h5ad	6.9	1.1		0.5	1.5	5.0	1.7	1.4	2.5	26		8.2
bixverse GPU*	6.1	0.7		0.3	0.5	2.1	0.9	1.0	0.6	24		4.4
Scanpy mtx	21	0.8	2.0	1.3	1.4	8.7	28	1.9	27	103		32
Scanpy h5ad	3.3	0.8	2.1	1.4	1.4	8.5	30	2.1	27	103		32
Scanpy dask	5.9	1.0	0.0	1.7	3.9	8.4	24	2.0	27	100		31
scrapper mtx	42	0.5	0.1	0.2	1.8	11	4.6	8.6	33	18	41	
scrapper h5ad	6.3	0.4	0.1	0.2	1.6	12	4.3	8.2	32	17	37	
Seurat std mtx	44	2.8	4.9	7.3	25	13	13	17	58	74		26
Seurat std h5ad	9.9	2.6	4.7	6.8	24	13	13	16	54	71		24
Seurat BPCells	14	1.2	2.0	2.6	73	14	14	18	56	74		24
OSCA HDF5	38	10	8.0	609	63	14	29	7.8	28	99		

500k (346,259 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	73	2.8		1.0	3.5	31	8.8	7.8	21	156		40
bixverse h5ad	31	3.3		1.1	3.5	33	9.2	7.9	21	154		40
bixverse GPU*	31	2.2		0.9	2.0	6.9	4.2	5.9	2.6	134		19
Scanpy mtx	OOM											
Scanpy h5ad	19	3.8	3.5	3.1	5.7	58	32	13	158	840		115
Scanpy dask	23	4.2	0.0	5.0	10	58	28	13	152	813		110
scrapper mtx	OOM											
scrapper h5ad	150	2.1	0.4	0.8	10	96	26	83	150	98	202	
Seurat std mtx	OOM											
Seurat std h5ad	OOM											
Seurat BPCells	72	4.7	8.3	8.8	358	92	94	113	340	639		108
OSCA HDF5	160	47	29	OOM (worker)								

full (684,932 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_matched	tsne_fft
bixverse mtx	148	4.5		1.8	6.3	65	17	16	48	373		121
bixverse h5ad	78	4.9		1.8	6.2	63	18	16	51	371		125
bixverse GPU*	74	2.9		1.8	5.6	16	8.3	12	6.0	325		40
Scanpy mtx	OOM											
Scanpy h5ad	OOM											
Scanpy dask	42	7.4	0.0	9.0	17	138	33	30	352	1,955		258
scrapper mtx	OOM											
scrapper h5ad	OOM											
Seurat std mtx	OOM											
Seurat std h5ad	OOM											
Seurat BPCells	143	9.5	16	16	702	213	201	250	760	1,645		229
OSCA HDF5	316	101	61	OOM (worker)								

wall time: 1 s, 10 s, 100 s, 1,000 s

bixverse CPU tSNE builds its kNN with IVF at 500k and full (kmknn at 100k).